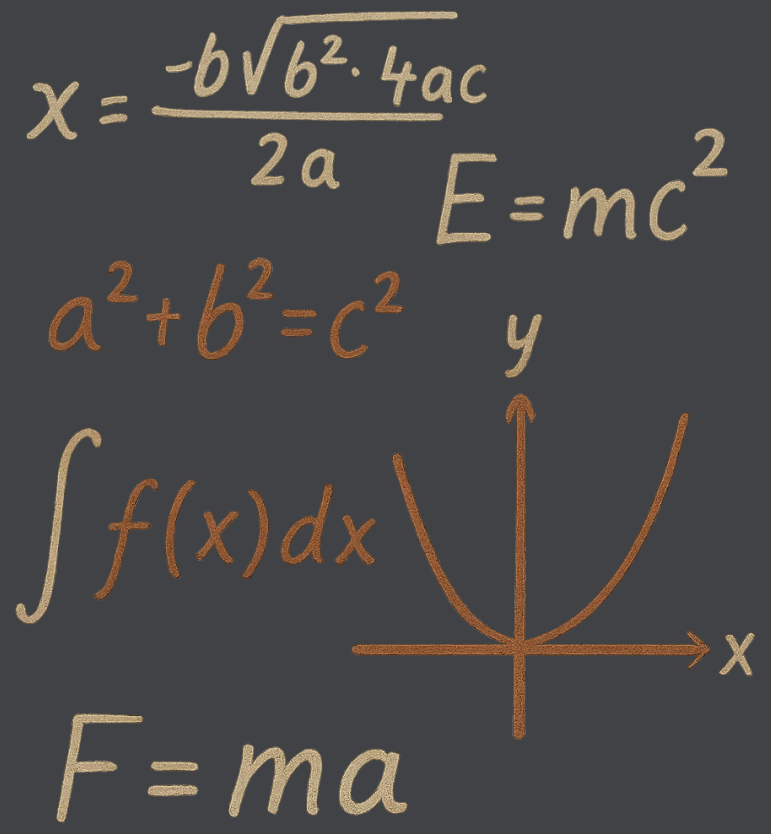
Calculus for B.Tech

Single Variable Applications

A comprehensive journey through the mathematical foundations that power modern engineering — from limits and derivatives to differential equations and series.

B.TECH MATHEMATICS

8 MODULES



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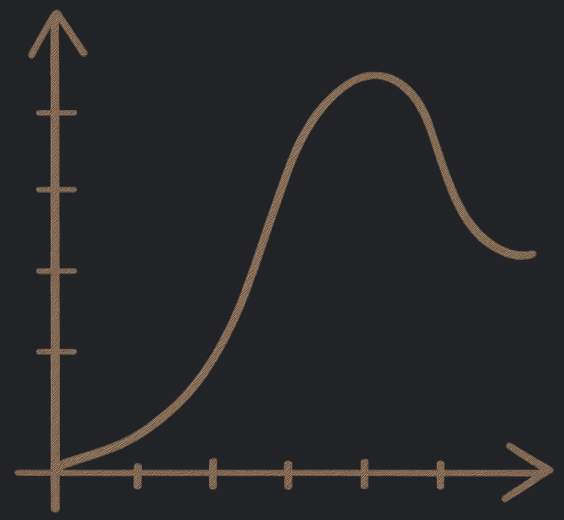
MODULE 1

Foundations of Calculus

Limits, Continuity, and Derivatives — the bedrock of all engineering mathematics.



Limits

Behaviour of a function as input approaches a value. Essential for defining derivatives.

Continuity

A function is continuous if small changes in input produce small changes in output — no breaks or jumps.

Derivatives

Instantaneous rate of change. Geometrically, the slope of the tangent line at a point.

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MODULE 2

Mastering Differentiation

Techniques and real-world engineering problems — from product rules to optimisation.

Product & Quotient Rule

Differentiate products and ratios of functions efficiently.

Chain Rule

Differentiate composite functions — critical for complex engineering models.

Optimisation

Find maxima and minima to design efficient structures and systems.

Implicit Differentiation

Handle equations where y is not explicitly expressed in terms of x.

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MODULE 3

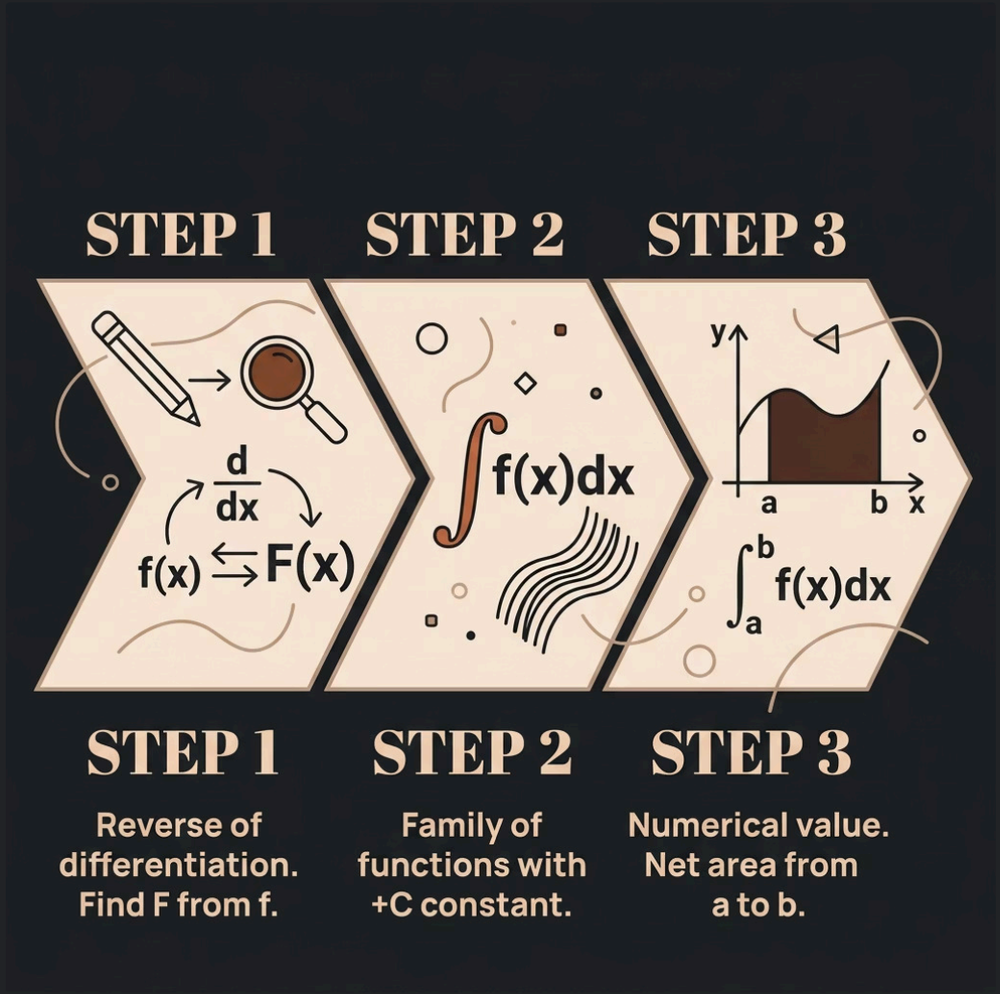
Introduction to Integration

Antiderivatives and the Fundamental Theorem of Calculus — reversing the derivative to accumulate change.

What is Integration?

Integration finds the accumulated quantity — area under a curve, total displacement, or total charge. The **Fundamental Theorem** links differentiation and integration as inverse operations.

If $F'(x) = f(x)$, then $\int_a^b f(x)\,dx = F(b) - F(a)$



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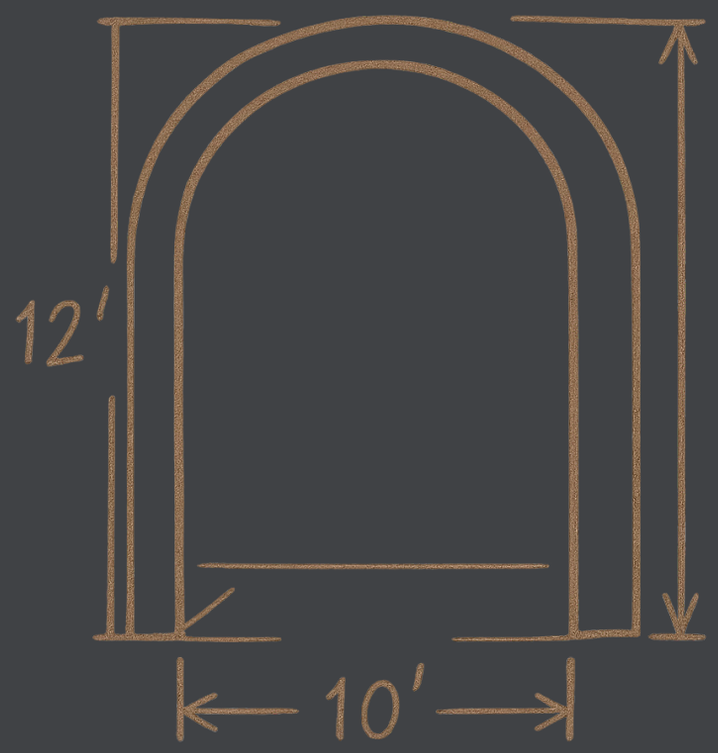

MODULE 4

Applications of Definite Integrals

Area, Volume, and Work Done — calculus in action across engineering disciplines.

1

Area Under a Curve

Compute irregular areas — used in fluid flow, load distributions, and signal processing.

2

Volume of Revolution

Disc and shell methods calculate volumes of tanks, pipes, and rotating machinery.

3

Work Done

$W = \int F(x)\,dx$ — calculate work by variable forces in mechanical systems.

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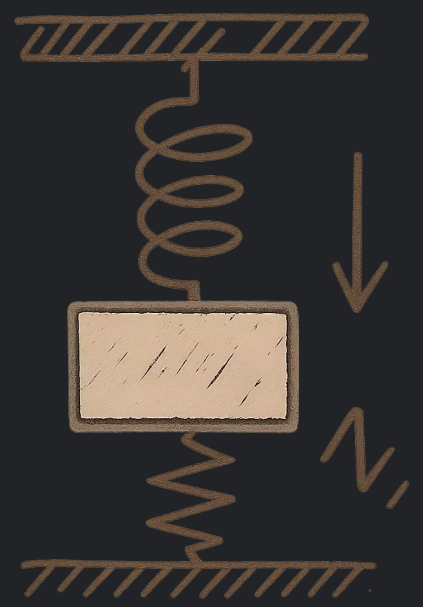
MODULE 5

Differential Equations in Engineering

Solving **First-Order ODEs** — the language of dynamic engineering systems.



Separable Equations

Variables separated on each side — used in growth/decay and cooling problems.

Linear First-Order ODEs

Solved using integrating factors — models RC circuits and mixing problems.

Engineering Applications

Population dynamics, radioactive decay, Newton's Law of Cooling, and electrical circuits.

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MODULE 6

Series and Sequences

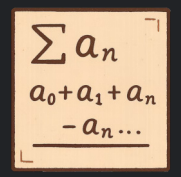
Convergence, Divergence, and Power Series — understanding infinite sums in engineering analysis.

Convergence

A series converges if its partial sums approach a finite limit. Tested using Ratio, Root, and Comparison tests.

Divergence

A series diverges when sums grow without bound or oscillate. The Divergence Test is the first check.



Power Series

Series of the form $\sum a_n x^n$. They represent functions within a radius of convergence — vital for approximations.

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MODULE 7

Taylor and Maclaurin Series

Approximating Functions using polynomials — a cornerstone of numerical engineering methods.

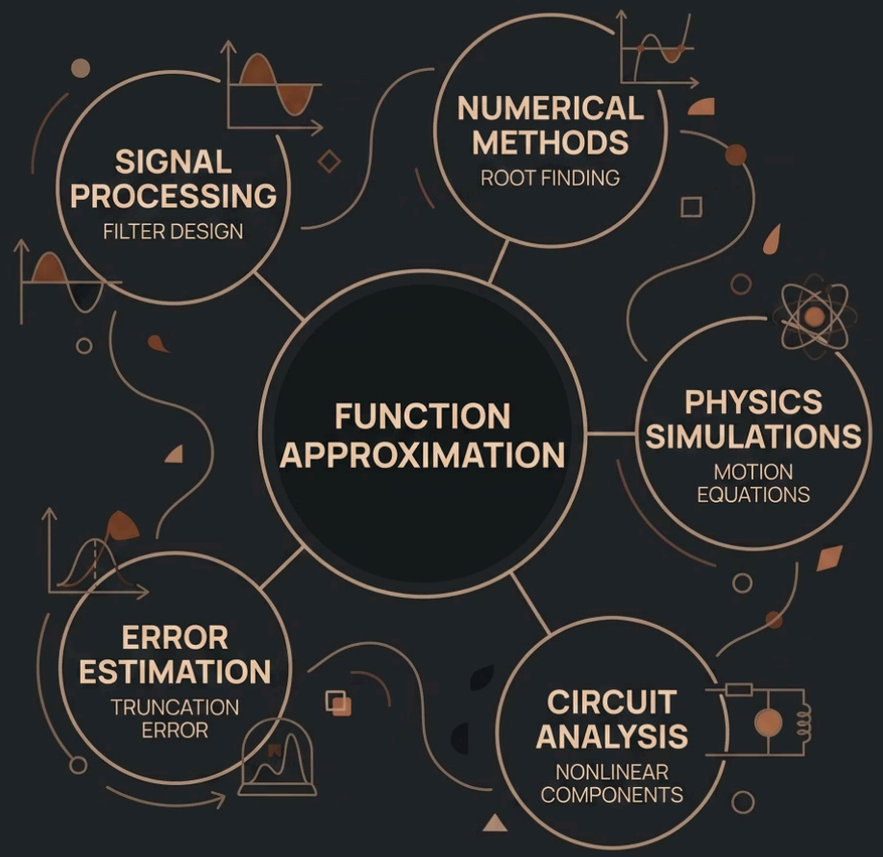
Taylor Series

Any smooth function can be approximated near a point $x = a$:

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots$$

Maclaurin Series

A special case where $a = 0$. Used for $\sin x$, $\cos x$, and e^x in circuit and vibration analysis.



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MODULE 8

Practical Applications of Single Variable Calculus

Case studies bridging theory and real-world engineering challenges.

🏗️ Civil Engineering

Optimising beam deflection curves and computing volumes of earthwork using integration.

⚡ Electrical Engineering

Solving RC and RL circuit ODEs to model transient current and voltage responses.

🔧 Mechanical Engineering

Analysing velocity, acceleration, and work-energy relationships in moving systems.

💻 Computer Engineering

Using Taylor series for function approximation in embedded systems and graphics rendering.

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SUMMARY

Key Takeaways & Future Scope

From concepts to engineering solutions — calculus is the language of change.

Core Concepts Mastered

01

Limits & Continuity

Foundation of all calculus operations.

02

Differentiation & Integration

Tools for rate of change and accumulation.

03

ODEs & Series

Models for dynamic systems and approximations.

Where to Next?

- Multivariable Calculus — gradients, divergence, curl
- Vector Calculus — line and surface integrals
- Laplace Transforms — solving higher-order ODEs
- Numerical Methods — computational calculus for simulations

Calculus is not just a subject — it is the framework behind every engineering innovation.

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